Type MS · Pigpen cipher

Answer 1 of 5 · Question 4 part b · 3 marks

2022 JUN · Paper 02 · extract 1 of 6 · find part b in this mark scheme extract

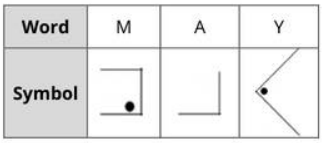
4 (a)	A1	**The only correct answer is B** A is not correct because this is decryption C is not correct because this is not an encryption D is not correct because this is not an encryption		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Pigpen cipher

Answer 1 of 5 · Question 4 part b · 3 marks

2022 JUN · Paper 02 · extract 2 of 6 · find part b in this mark scheme extract

4 (b)	B1 B2 B3	**Award 1 mark for each correct symbol**  		(3)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Pigpen cipher

Answer 1 of 5 · Question 4 part b · 3 marks

2022 JUN · Paper 02 · extract 3 of 6 · find part b in this mark scheme extract

4 (c)(i)	C1	5		(1)

Type MS · Pigpen cipher

Answer 1 of 5 · Question 4 part b · 3 marks

2022 JUN · Paper 02 · extract 4 of 6 · find part b in this mark scheme extract



Type MS · Pigpen cipher

Answer 1 of 5 · Question 4 part b · 3 marks

2022 JUN · Paper 02 · extract 5 of 6 · find part b in this mark scheme extract

4 (c)(iii)	C3	Rail (Fence cipher)		(1)

Type MS · Pigpen cipher

Answer 1 of 5 · Question 4 part b · 3 marks

2022 JUN · Paper 02 · extract 6 of 6 · find part b in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Pigpen cipher

Answer 2 of 5 · Question 4 part a · 1 mark

2019 JUN · Paper 02 · extract 1 of 4 · find part a in this mark scheme extract

4(a)	**The only correct answer is D** A is not correct because Plaintext is the original unencrypted text B is not correct because there is a consistency in the coding of the text C is not correct because a shift encryption replaces letters with other letters a given distance from the original letter (as in a Caesar cipher)		1

Type MS · Pigpen cipher

Answer 2 of 5 · Question 4 part a · 1 mark

2019 JUN · Paper 02 · extract 2 of 4 · find part a in this mark scheme extract

4(b)	Award 3 marks for all rows correct Award 2 marks for 4 rows correct Award 1 mark for 2 rows correct G Q I X N U		(3)

Type MS · Pigpen cipher

Answer 2 of 5 · Question 4 part a · 1 mark

2019 JUN · Paper 02 · extract 3 of 4 · find part a in this mark scheme extract

Q4(c)	One mark for correct decision: Yes (1) Up to two marks for the correct justifications: Each/every letter has a symbol / is represented (1) The symbol is a single/unique representation (1)	Award marks independently	(3)

Type MS · Pigpen cipher

Answer 2 of 5 · Question 4 part a · 1 mark

2019 JUN · Paper 02 · extract 4 of 4 · find part a in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Pigpen cipher

Answer 3 of 5 · Question 4 part b · 3 marks

2019 JUN · Paper 02 · extract 1 of 4 · find part b in this mark scheme extract

4(a)	**The only correct answer is D** A is not correct because Plaintext is the original unencrypted text B is not correct because there is a consistency in the coding of the text C is not correct because a shift encryption replaces letters with other letters a given distance from the original letter (as in a Caesar cipher)		1

Type MS · Pigpen cipher

Answer 3 of 5 · Question 4 part b · 3 marks

2019 JUN · Paper 02 · extract 2 of 4 · find part b in this mark scheme extract

4(b)	Award 3 marks for all rows correct Award 2 marks for 4 rows correct Award 1 mark for 2 rows correct G Q I X N U		(3)

Type MS · Pigpen cipher

Answer 3 of 5 · Question 4 part b · 3 marks

2019 JUN · Paper 02 · extract 3 of 4 · find part b in this mark scheme extract

Q4(c)	One mark for correct decision: Yes (1) Up to two marks for the correct justifications: Each/every letter has a symbol / is represented (1) The symbol is a single/unique representation (1)	Award marks independently	(3)

Type MS · Pigpen cipher

Answer 3 of 5 · Question 4 part b · 3 marks

2019 JUN · Paper 02 · extract 4 of 4 · find part b in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Pigpen cipher

Answer 4 of 5 · Question 4 part c · 3 marks

2019 JUN · Paper 02 · extract 1 of 4 · find part c in this mark scheme extract

4(a)	**The only correct answer is D** A is not correct because Plaintext is the original unencrypted text B is not correct because there is a consistency in the coding of the text C is not correct because a shift encryption replaces letters with other letters a given distance from the original letter (as in a Caesar cipher)		1

Type MS · Pigpen cipher

Answer 4 of 5 · Question 4 part c · 3 marks

2019 JUN · Paper 02 · extract 2 of 4 · find part c in this mark scheme extract

4(b)	Award 3 marks for all rows correct Award 2 marks for 4 rows correct Award 1 mark for 2 rows correct G Q I X N U		(3)

Type MS · Pigpen cipher

Answer 4 of 5 · Question 4 part c · 3 marks

2019 JUN · Paper 02 · extract 3 of 4 · find part c in this mark scheme extract

Q4(c)	One mark for correct decision: Yes (1) Up to two marks for the correct justifications: Each/every letter has a symbol / is represented (1) The symbol is a single/unique representation (1)	Award marks independently	(3)

Type MS · Pigpen cipher

Answer 4 of 5 · Question 4 part c · 3 marks

2019 JUN · Paper 02 · extract 4 of 4 · find part c in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Pigpen cipher

Answer 5 of 5 · Question 3 part c · 2 marks

2025 JUN · Paper 01 · extract 1 of 3 · find part c in this mark scheme extract

3(a)(i)	Award **one** mark for each correct hexadecimal character in the correct order: • B9		2
Question	Answer	Additional Guidance	Mark
3(a)(ii)	**The only correct answer is B** *A is not correct because both are processed at the same speed by a computer* *C is not correct because both use the same amount of storage* *D is not correct because hexadecimal uses number base 16*		1
Question	Answer	Additional Guidance	Mark
3(a)(iii)	Award **two** marks for a linked description such as: • (logical/arithmetic) right shift (1) by 2 places (1)		2
Question	Answer	Additional Guidance	Mark
3(b)(i)	Award **one** mark for: Pixel/picture element (1)	Accept additional words. Accept incorrect spelling if word is recognisable as pixel	1

Type MS · Pigpen cipher

Answer 5 of 5 · Question 3 part c · 2 marks

2025 JUN · Paper 01 · extract 2 of 3 · find part c in this mark scheme extract

Question	Answer	Additional Guidance	Mark
3(b)(ii)	Award **one** mark for each of: • 16x16 pixels (1) • Multiplying (16x16) by 2 bits per pixel in the numerator for colour depth (1) • Dividing by 8 in the denominator to convert to bytes (1) • Dividing 1000 by the number of bytes used for one image (1) Example: $1000 / \frac{16 \times 16 \times 2}{8}$ bytes Allow follow-through	4 marks for any of: 1000 X 8 / 16 x 16 x 2 8000 / 256 x 2 1000 / 16 x 2 x 2 1000 / 16 x 4 8000 / 512	4
Question	Answer	Additional Guidance	Mark
3(b)(iii)	Award up to **two** marks for a linked explanation such as: • high compression ratio (1) repeated/continuous runs (of the same pixel colour) (1) • requires less storage space (1) encoding 16W for line 1 (1)	Allow valid RLE examples from the image. Explanation can be a combination of different marking points. E.g. MP1a and MP2b	2
Question	Answer	Additional Guidance	Mark
3(c)	Award **one** mark for each correct cell: Encrypted text / Plain text  RED  COT	Ignore capitalisation Must be correct spelling	2
		Total for question 3	**14**

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · Pigpen cipher

Answer 5 of 5 · Question 3 part c · 2 marks

2025 JUN · Paper 01 · extract 3 of 3 · find part c in this mark scheme extract

Question	Answer	Additional Guidance	Mark